An Adaptive High-Radix and Parametric Troposcatter Communication System

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Title: An Adaptive High-Radix and Parametric Troposcatter Communication System

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Cross-Reference to Related Applications

This application is related to the inventions described in the U.S. Patent Applications for "A Method and System for High-Radix Physical Layer Data Encoding and Representation" and "A Protocol Stack and System for Parametric Data Networking." The present invention applies these core principles to the uniquely challenging and dynamic environment of troposcatter communication, the contents of which are hereby incorporated by reference in their entirety.

Abstract

An adaptive troposcatter communication system for transmitting high-radix, parametric data over variable beyond-line-of-sight (BLOS) radio channels. The system comprises adaptive transceivers that continuously monitor the troposcatter channel's conditions. Based on this real-time analysis, the transceivers dynamically adjust a symbol-to-state mapping that defines a set of N discrete, distinguishable RF states (where N greater than 2) used to encode abstract symbols. The system can vary the radix (N) of the transmission, reducing it to

enhance reliability during poor channel conditions and increasing it to maximize throughput when conditions are favorable. Data is transported as unevaluated parametric expressions, ensuring end-to-end data fidelity is maintained across the inherently unstable troposcatter link.

Background of the Invention

Tropospheric scatter (troposcatter) communication provides a valuable capability for establishing beyond-line-of-sight (BLOS) links, particularly in military and remote operational contexts. However, the tropospheric channel is characterized by severe and rapid fading, multipath distortion, and high path loss, making it notoriously difficult. Conventional troposcatter systems focus on achieving a robust *binary* data link through powerful error correction, diversity schemes, and complex modulation techniques. These systems are fundamentally limited to transporting bit-streams, meaning any high-precision or symbolic data must first be converted to a lossy, approximated binary format before transmission.

Summary of the Invention

The present invention defines an adaptive troposcatter communication system that natively transmits high-radix symbolic data, overcoming the limitations of binary systems and preserving perfect data precision across the link. The system's core innovation is its ability to adapt its physical layer encoding to the volatile nature of the troposcatter channel.

The system comprises a pair of adaptive transceivers. Each transceiver includes a channel condition monitoring unit that continuously analyzes key channel metrics (e.g., signal-to-noise ratio, fading characteristics). A dynamic symbol-to-state mapper uses this real-time data to define and continuously update an optimal set of N distinguishable RF states for transmission. The number of states, N (the radix), is not fixed; the mapper can increase N when the channel is clear to boost data rate, and decrease N to a smaller, more robust set of states when the channel is degraded, thereby ensuring link reliability.

The system's modulator directly converts abstract symbols (representing numbers, operators, or other data primitives) into this dynamically defined set of RF states. Data is encapsulated in parametric packets, allowing unevaluated mathematical "recipes" to be transmitted as the payload. This ensures that even when the radix N is dynamically changed, the symbolic integrity of the underlying data is preserved from end to end.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating the adaptive high-radix parametric data flow in a troposcatter communication system.

FIG. 2 is a block diagram of an adaptive troposcatter transceiver highlighting the dynamic symbol-to-state mapping based on channel monitoring.

Detailed Description of the Invention

Referring to **FIG. 1**, the adaptive troposcatter system (100) is shown. A first adaptive transceiver (130) receives a parametric packet (120) containing a symbolic data structure (110). A channel monitoring unit within the transceiver assesses the volatile troposcatter path (140). Based on this assessment, an adaptive mapper (135) selects an optimal set of N RF states for encoding. The transceiver then transmits a high-radix RF signal (145) which is scattered by the troposphere. A second adaptive transceiver (150) receives the scattered signal. Its own adaptive mapper (155), using its own channel assessment, decodes the signal by selecting the correct inverse mapping. This allows the reconstruction of the original parametric packet (120), delivering the perfect-precision symbolic data (110) to the destination.

FIG. 2 provides a block diagram of the adaptive transceiver (200). A Channel Condition Monitor (210) continuously analyzes the received signal from the antenna (250). It sends metrics to a Dynamic Symbol-to-State Mapper (220). This mapper defines the optimal radix N and the specific mapping of symbols to RF states. It provides this map to both the High-Radix RF Modulator (230), which encodes outgoing abstract symbols, and the High-Radix

RF Demodulator (240), which decodes incoming RF signals. This closed-loop adaptation ensures the most efficient and reliable radix is used at all times.

Claims

What is claimed is:

1. An adaptive troposcatter communication system, comprising:

    - a first adaptive transceiver, comprising:
        - a channel condition monitor configured to assess real-time characteristics of a troposcatter communication channel;
        - a dynamic symbol-to-state mapper configured to, based on the assessed channel characteristics, dynamically define a set of N discrete, distinguishable radio frequency (RF) states, where N is an integer greater than two, and establish a mapping between abstract symbols and said N RF states;
        - a high-radix RF modulator configured to receive a sequence of said abstract symbols and generate a corresponding sequence of said N RF states; and
    - a second adaptive transceiver, communicatively coupled to the first transceiver via the troposcatter communication channel, the second adaptive transceiver comprising a high-radix RF demodulator configured to, based on an inverse of the mapping, decode a received sequence of said N RF states into a sequence of abstract symbols.

2. The system of claim 1, wherein the dynamic symbol-to-state mapper is configured to adjust the value of N based on the channel conditions.

3. The system of claim 2, wherein the mapper is configured to reduce N during periods of channel degradation to enhance reliability.

4. The system of claim 2, wherein the mapper is configured to increase N during periods of favorable channel conditions to enhance data throughput.

5. The system of claim 1, wherein the abstract symbols are components of a parametric data packet, the payload of which is an unevaluated mathematical expression.

6. A method for adaptive troposcatter communication, comprising:

- continuously assessing real-time characteristics of a troposcatter communication channel;
- dynamically defining a set of N discrete, distinguishable radio frequency (RF) states based on said assessed channel characteristics, where N is an integer greater than two;
- establishing a mapping between abstract symbols and said N RF states;
- encoding a sequence of said abstract symbols into a corresponding sequence of said N RF states;
- transmitting said sequence of RF states via a troposcatter link; and
- receiving and decoding said sequence of RF states back into the original sequence of abstract symbols using an inverse of the mapping.

7. The method of claim 6, wherein the value of N is dynamically adjusted based on the assessed channel characteristics.

8. The method of claim 6, wherein the abstract symbols represent an unevaluated mathematical expression that is transported across the troposcatter link without loss of precision.

Drawings

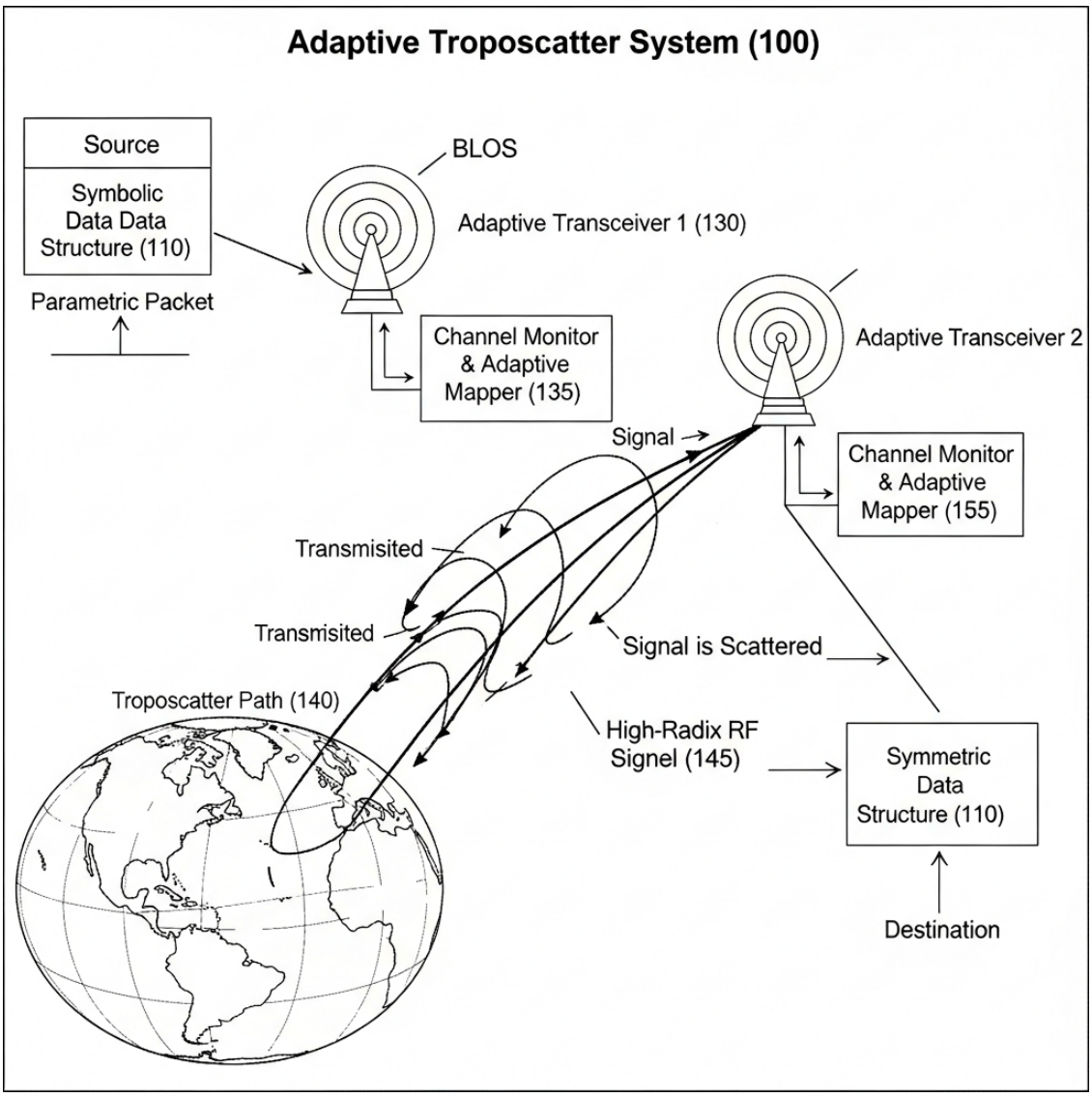


Figure 1: Conceptual diagram illustrating the adaptive high-radix parametric data flow in a troposcatter communication system.

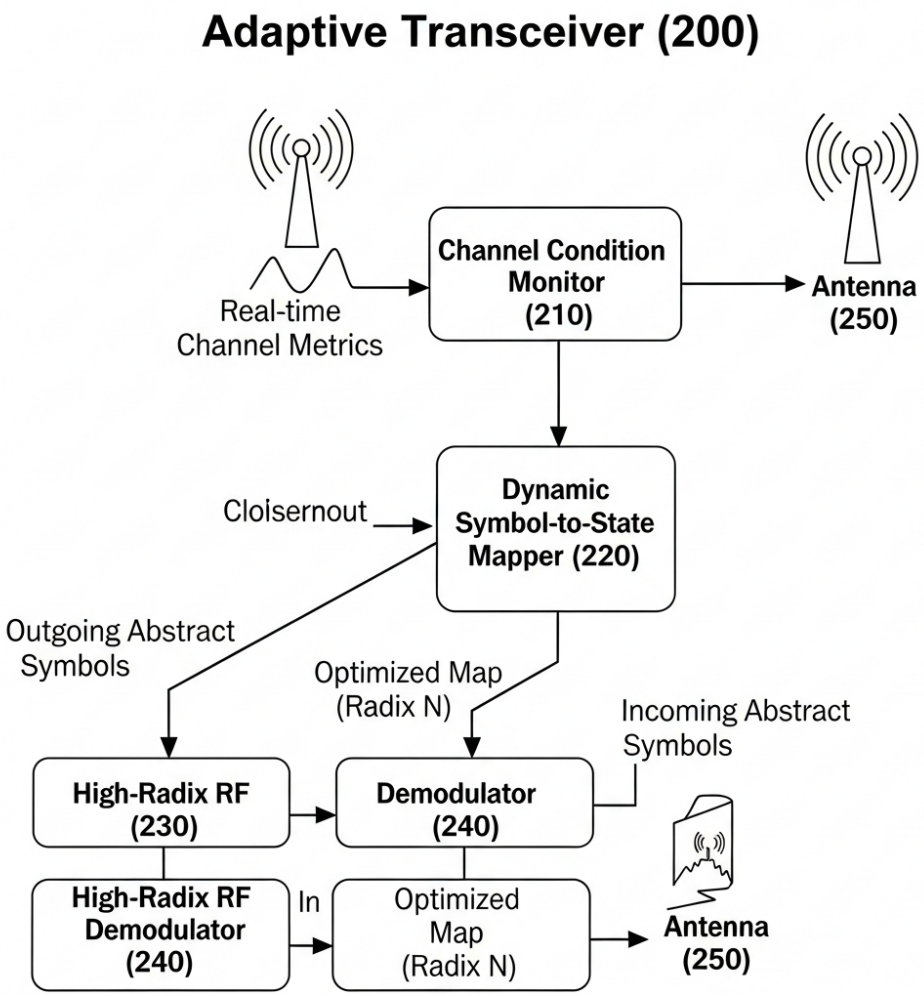


Figure 2: Block diagram of an adaptive troposcatter transceiver showing the dynamic symbol-to-state mapping.